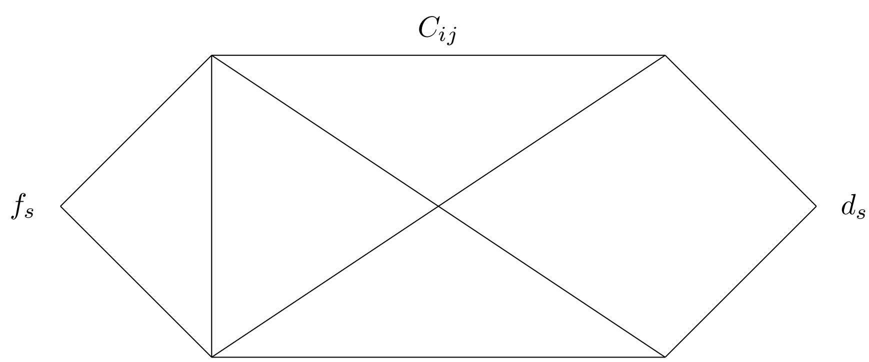
ECE 647: Lecture 19

Routing

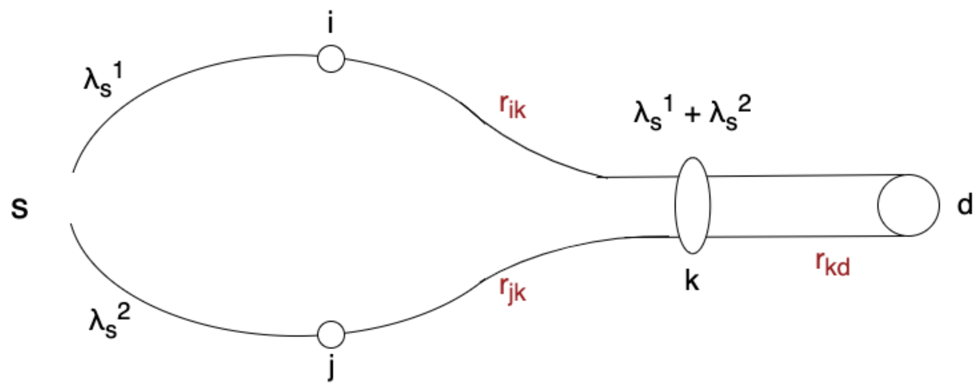


- S users
- Each user $s = 1, 2, ..., S$ sends a flow from node f_s to d_s at the rate of X_s
- L: set of links
  - each link (i, j) from node i to node j
  - capacity of link (i, j) is C_{ij}

Q: How to route the flows?

Basic Node-Balance Equation

- Let r_{ij}^s denote the amount of capacity on link (i, j) that is allocated to flows



- Then any feasible flow (routing) is equivalent to $[r_{ij}^s]$ that satisfies the following node balance equation:

For any node i:

$$(*) \quad \sum_{j:(i,j)\in L} r_{ij}^s + \sum_s X_s 1_{\{i=d_s\}} \rightarrow \text{total outgoing flow at node } i$$

$$= \sum_{j:(j,i)\in L} r_{ji}^s + \sum_s X_s 1_{\{i=f_s\}} \rightarrow \text{total incoming flow at node } i$$

Capacity Constraints

$$(**) \ \sum_s r_{ij}^s \leq C_{ij} \text{ (if link capacity on each direction is separate)}$$

$$or \ \sum_s r_{ij}^s + r_{ji}^s \leq C_{ij} = C_{ji} \text{ (if a whole bi-direction capacity is defined)}$$

Objectives

ⓐ Maximize Utility

$$\max \sum_s U_s(X_s)$$

sub. to. $(*)$ & $(**)$

Other possibilities:

(b) Just feasibility

$$\min 0$$

$$\text{sub. to. } (*) \ \& \ (**)$$

(c) Maximize future throughput (in proportion to a reference λ_s)

- Find the largest α such that the rate of each flow can be increased to $\alpha\lambda_s$

$$\max \alpha$$

$$\text{sub. to. } \lambda_s = \alpha\lambda_s \quad \forall \quad s$$

$$(*) \ \& \ (**)$$

- When there is only one commodity, it reduces to the max-flow problem

(d) Minimize congestion at fixed $X_s = \lambda_s$

- Define a congestion measure for each link $\beta_{ij}(\sum_s r_{ij}^s)$
- Minimize the total congestion level of the network
- Homework: formulate convex problems

Ref: Ch 5. Bertsekas Tsitsiklis

Parallel Distributed Computation:

Numerical Methods.

Cross Layer

So far all these examples easily lead to convex problems

- The capacity of each link is assumed to be fixed

We will see that this changes quickly when we allow the link rate to be a function of other control variables

- This is common in wireless networks
- The link rate will depend on
  - Transmission power
  - Transmission schedule

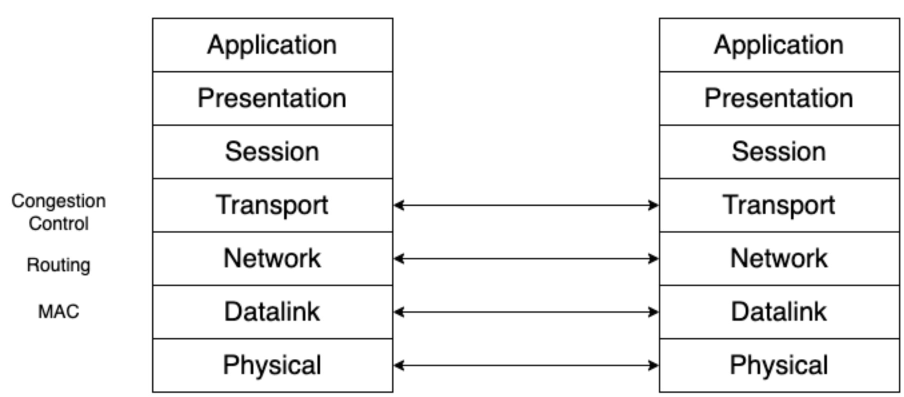
We may still formulate these control decisions into a unified optimization problem

→ "Cross-layer control"

- rate-control → Transport layer
- routing → Network layer
- link scheduling → MAC layer
- power control → Physical layer

Q. Why do we want to consider multiple layers together?

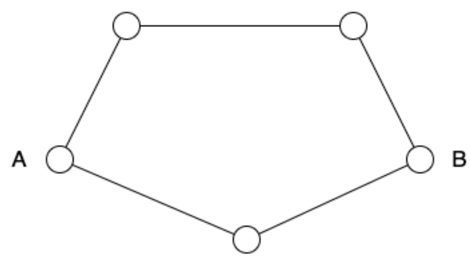
- In wireless networks, often the protocols are classified into layers
- Layering is a form of hierarchical modularity
- The higher layer use the service provided by the lower layer. But it does not need to know the inner working of the lower layer



- Benefits of Modularity
  - Easy to understand
  - Easy to change
- However for some wireless networks, examples have been found where such a layering architecture can limit performance

Example

- Typically, routing is designed to minimize the number of hops



- Tend to use "long" links
- In wireless networks, long transmission link can suffer from a low SINR
  - poor end-to-end performance
- It would be better if the routing protocol takes into account the physical layer characteristics
- Pitfalls of Cross-layer Design
  - Loss of modularity
  - Fragile solution that is hard to change
  - Would be highly desirable if we can still obtain "modular" solutions
  - Need duality/decomposition